Ezequiel T. **Centofanti**

AI Research Engineer · Data Scientist

Paris, France

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Summary

PhD Candidate specialised in AI for Astrophysics with a strong background in Machine Learning, Deep Learning, and Signal Processing. Experienced in developing novel algorithms for inverse problems and processing complex scientific data. Proficient in Python (PyTorch, TensorFlow, JAX) and High-Performance Computing (HPC), with a proven ability to bridge the gap between theoretical research and scalable solutions.

Experience

CEA & Université Paris Cité — *Paris, France*

PhD Researcher Machine Learning and AI for Astrophysics — *2023 - Present*

- Designed and trained a **ML classifier** for stellar spectral types, incorporating the telescope's instrumental response; improved classification accuracy from $75\% \rightarrow 90\%$.
- Developed a lightweight, modular Python package for radio-interferometric simulations, following **industry-grade software** engineering standards: unit tests, full documentation, CI/CD, Docker, PyPI packaging.
- Proposed a novel training strategy for a data-driven PSF model enabling phase-retrieval, achieving a $10\times$ reduction in wavefront error.
- Implemented a **generative forward model** for galaxy shear estimation, demonstrating unbiased predictions with lower variance than state-of-the-art CNN approaches, improving precision for weak-lensing measurements in next-generation surveys.

Inria (OPIS Group) — *Paris, France*

AI Research Engineer — *2022 - 2023*

- Built scalable **Deep Unrolling** solutions for inverse imaging with **PyTorch** on **HPC**, delivering fully reproducible open-source implementations.

CosmoStat, CEA — *Paris, France*

Research Intern (M2) — *2022*

- Point Spread Function (PSF) estimation model using **TensorFlow** for the European Space Agency's *Euclid* telescope.

Thales Group & IMT Atlantique — *Brest, France*

Engineering Intern (M1) — *2021*

- Extracted micro-Doppler features for UAV detection/classification, using Matlab and Python.

Polytech Paris-Saclay — *Paris, France*

Assistant Professor (Digital & Analog Electronics) — *2024 - 2025*

- Teaching practical lab sessions on digital and analogue electronics to engineering students.

Education

CEA & Université Paris Cité — *Paris, France*

PhD in Machine Learning & AI for Astronomical Imaging — *2023 - Present*

- Advisors: Dr. Samuel Farrens & Dr. Jean-Luc Starck.
- Research Focus: Developing AI methods for optical and radio astronomical imaging.

IMT Atlantique (Co-accredited with Univ. Rennes 1 & CentraleSupélec) — *Brest, France*

Diplôme d'Ingénieur — *2022*

- Specialised Master (SISEA) in systems, automation, and advanced signal processing.
- M2: Mathematical and Computational Engineering: Deep Learning, Computer Vision, Optimization, and Numerical Methods.
- M1: Embedded Systems, FPGA programming, and Circuit Design.

University of Buenos Aires — *Buenos Aires, Argentina*

Electrical Engineering Degree — *2019*

- Focus on Analog/Digital Electronics, Signal Processing, and Control Systems.

Technical Skills

Machine Learning	Deep Learning, Signal Processing, Inverse Problems, Optimization, Generative Models, Wheights & Biases
Languages & Frameworks	Python (PyTorch, Scikit-Learn, NumPy, Matplotlib, pytest, PyPI packaging), JAX, C/C++, AI Coding Agents
Statistical Methods	Probabilistic programming languages (numpyro, blackjax), Bayesian inference (emcee, MCMC, HMC)
Tools & Platforms	Git, GitHub Actions, High-Performance Computing (HPC), Linux/Unix, Docker, LaTeX
Languages	English (Fluent), French (Fluent), Spanish (Native), Italian (Intermediate)

Honors & Awards

2023	**Distinguished Student**, University of Buenos Aires	*Argentina*
2021	**Academic Distinction**, International Double Degree Program	*France/Argentina*
2020	**EIFFEL Excellence Scholarship**, Awarded by the French Ministry of Foreign Affairs	*France*